Data Science Target 2 Improved

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r Sys.Date()

Setup chunk - configure global options and load required libraries

Data Import

```
# Import the CSV file containing youth NEET (Not in Education, Employment, or Training) data
youth_not_eet <- read.csv("/Users/vasilikimichalia/Desktop/IDS_Group_Proj-main/Exported_data/youth_not_

# Rename the fourth column to "Percentage" for clearer interpretation
colnames(youth_not_eet)[4] <- "Percentage"
```

Analysis and Visualization

```
# STEP 1: Calculate weighted averages by continent and year
# This accounts for population differences between countries
youth_summary <- youth_not_eet %>%
  group_by(Continent, Year) %>%
  # Use population as weight to give larger countries more influence
  summarise(weighted_average = weighted.mean(Percentage, Population, na.rm = TRUE),
            .groups = "drop") %>%
  ungroup()

# STEP 2: Define color scheme for continents
# Using consistent colors for visual clarity across all plots
continent_colors <- c(
  "Africa"        = "#F8766D",  # Red
  "Asia"          = "#B79F00",  # Gold
  "Europe"        = "#00BA38",  # Green
  "North America" = "#00BFC4",  # Cyan
  "Oceania"       = "#619CFF",  # Blue
  "South America" = "#F564E3"   # Pink
)

# STEP 3: Define outlier removal function
# Uses Interquartile Range (IQR) method with aggressive 1.0 multiplier
remove_outliers <- function(data, variable) {
  # Calculate first quartile (25th percentile)
  Q1 <- quantile(data[[variable]], 0.25, na.rm = TRUE)
  # Calculate third quartile (75th percentile)
  Q3 <- quantile(data[[variable]], 0.75, na.rm = TRUE)
  # Calculate interquartile range
```

```r
  IQR_val <- Q3 - Q1

  # Define bounds: values outside these are considered outliers
  # Using 1.0 multiplier (aggressive) instead of typical 1.5
  lower_bound <- Q1 - 1.0 * IQR_val
  upper_bound <- Q3 + 1.0 * IQR_val

  # Filter data to keep only values within bounds
  data %>% filter(
    !!sym(variable) >= lower_bound &
      !!sym(variable) <= upper_bound
  )
}

# STEP 4: Apply outlier removal to create cleaned dataset
youth_summary_no_outliers <- youth_summary %>%
  group_by(Continent) %>%
  remove_outliers("weighted_average") %>%
  ungroup()

# STEP 5: Calculate regression slopes WITH outliers
# This shows the original trend before data cleaning
slope_with <- youth_summary %>%
  group_by(Continent) %>%
  do({
    # Fit linear model: NEET percentage vs Year
    model <- lm(weighted_average ~ Year, data = .)
    # Convert model to tidy format
    tidy_model <- tidy(model)
    # Extract the Year coefficient (slope)
    slope_row <- tidy_model %>% filter(term == "Year")

    # Create result data frame with slope and p-value
    data.frame(
      Continent = unique(.$Continent),
      slope_with = slope_row$estimate,      # Rate of change per year
      p_with = slope_row$p.value,           # Statistical significance
      stringsAsFactors = FALSE
    )
  }) %>%
  ungroup()

# STEP 6: Calculate regression slopes WITHOUT outliers
# This shows the trend after removing extreme values
slope_without <- youth_summary_no_outliers %>%
  group_by(Continent) %>%
  do({
    # Fit linear model on cleaned data
    model <- lm(weighted_average ~ Year, data = .)
    tidy_model <- tidy(model)
    slope_row <- tidy_model %>% filter(term == "Year")

    # Create result data frame
```

```r
    data.frame(
      Continent = unique(.$Continent),
      slope_without = slope_row$estimate,
      p_without = slope_row$p.value,
      stringsAsFactors = FALSE
    )
  }) %>%
  ungroup()

# STEP 7: Create comprehensive comparison table
# Combines results from both analyses (with and without outliers)
slope_comparison <- slope_with %>%
  left_join(slope_without, by = "Continent") %>%
  mutate(
    # Binary significance at 5% level
    sig_with = ifelse(p_with < 0.05, "Yes", "No"),
    sig_without = ifelse(p_without < 0.05, "Yes", "No"),

    # Star notation for significance levels (standard in academic reporting)
    sig_label_with = case_when(
      p_with < 0.001 ~ "***",   # Highly significant
      p_with < 0.01 ~ "**",     # Very significant
      p_with < 0.05 ~ "*",      # Significant
      TRUE ~ "n.s."             # Not significant
    ),
    sig_label_without = case_when(
      p_without < 0.001 ~ "***",
      p_without < 0.01 ~ "**",
      p_without < 0.05 ~ "*",
      TRUE ~ "n.s."
    ),

    # Track how outlier removal affected statistical significance
    significance_change = case_when(
      sig_with == "Yes" & sig_without == "No" ~ "Lost significance",
      sig_with == "No" & sig_without == "Yes" ~ "Gained significance",
      TRUE ~ "No change"
    ),

    # Calculate percentage change in slope estimate
    slope_change_pct = ((slope_without - slope_with) / abs(slope_with)) * 100
  )

# STEP 8: Display comparison table
# Shows side-by-side results for hypothesis testing
kable(
  slope_comparison %>%
    select(
      Continent,
      `Slope (with)` = slope_with,
      `P-value (with)` = p_with,
      `Sig (with)` = sig_label_with,
      `Slope (without)` = slope_without,
```

```
      `P-value (without)` = p_without,
      `Sig (without)` = sig_label_without,
      `Significant at 5%? (with)` = sig_with,
      `Significant at 5%? (without)` = sig_without,
      `Change in Significance` = significance_change
    ),
  digits = 4,
  caption = "Hypothesis Testing Results: With and Without Outliers (IQR multiplier: 1.0)"
)
```

Table 1: Hypothesis Testing Results: With and Without Outliers (IQR multiplier: 1.0)

Continent	Slope (with)	P-value (with)	Sig (with)	Slope (without)	P-value (without)	Sig (without)	Significant at 5%? (with)	Significant at 5%? (without)	Change in Significance
Africa	-0.5262	0.5288	n.s.	-0.5262	0.5288	n.s.	No	No	No change
Asia	1.5973	0.0060	**	1.5973	0.0060	**	Yes	Yes	No change
Europe	-0.3174	0.0368	*	-0.3174	0.0368	*	Yes	Yes	No change
North America	-0.0348	0.8649	n.s.	-0.0348	0.8649	n.s.	No	No	No change
Oceania	0.8516	0.0834	n.s.	0.8516	0.0834	n.s.	No	No	No change
South America	0.2993	0.1892	n.s.	0.2993	0.1892	n.s.	No	No	No change

```
# STEP 9: Identify which data points were removed as outliers
# Uses anti_join to find rows in original data not in cleaned data
outlier_rows <- youth_summary %>%
  anti_join(
    youth_summary_no_outliers,
    by = c("Continent", "Year", "weighted_average")
  )

# STEP 10: Display outlier information
if (nrow(outlier_rows) > 0) {
  # Show detailed list of removed data points
  kable(outlier_rows %>% arrange(Continent, Year),
        digits = 3,
        caption = "Data Points Removed as Outliers")

  # Create summary showing how many outliers per continent
  outlier_summary <- outlier_rows %>%
    group_by(Continent) %>%
    summarise(
      `Outliers Removed` = n(),
      `Years Affected` = paste(sort(Year), collapse = ", ")
    )
```

```r
  kable(outlier_summary, caption = "Outlier Summary by Continent")
} else {
  # If no outliers found, print message
  cat("No outliers detected using IQR method (multiplier: 1.0).\n")
}
```

```
## No outliers detected using IQR method (multiplier: 1.0).
```

```r
# STEP 11: Prepare slope annotations for visualization
# Creates formatted labels showing slope and significance
slope_data <- slope_comparison %>%
  select(Continent, slope_with, sig_label_with) %>%
  mutate(
    slope_label = paste0(
      "Slope: ", sprintf("%.3f", slope_with),  # Format to 3 decimal places
      " (", sig_label_with, ")"                # Add significance stars
    )
  )

# STEP 12: Create comprehensive visualization
# Shows trends for all continents with regression lines
combined_plot <- ggplot(youth_summary, aes(x = Year, y = weighted_average)) +
  # Add scatter points for actual data
  geom_point(aes(colour = Continent), alpha = 0.7, size = 3) +
  # Add linear regression trend line with confidence interval
  geom_smooth(aes(colour = Continent, fill = Continent),
              method = "lm", se = TRUE, linewidth = 1) +
  # Add slope labels in top-left of each facet
  geom_text(data = slope_data,
            aes(label = slope_label, colour = Continent),
            x = -Inf, y = Inf, hjust = -0.1, vjust = 1.2,
            size = 3.5, fontface = "bold") +
  # Apply custom color scheme
  scale_colour_manual(values = continent_colors) +
  scale_fill_manual(values = continent_colors) +
  # Create separate panel for each continent (free y-axis for clarity)
  facet_wrap(~ Continent, scales = "free_y") +
  # Add titles and axis labels
  labs(
    title = "Youth Not in EET: Trend by Continent",
    subtitle = "Weighted averages with linear regression trend lines",
    x = "Year",
    y = "NEET (%)"
  ) +
  # Apply minimal theme for clean appearance
  theme_minimal() +
  theme(
    legend.position = "none",  # Legend not needed due to faceting
    plot.title = element_text(size = 14, face = "bold"),
    strip.text = element_text(size = 12, face = "bold")
  )

# Display the plot
combined_plot
```

```
## `geom_smooth()` using formula = 'y ~ x'
```